Project Development Phase
Performance Test

Date	29 October 2026
Team ID	Individual Project — Ujjawal Rathod
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	

Project Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	ev_market_train.csv (200,000 rows) + ev_market_test.csv (100,000 rows) unioned into one data source: 300,000 rows × 37 columns, 15 vehicle categories, 200 manufacturers, 7 market regions, 10 manufacturing countries, model years 2018–2026. [Insert screenshot: Data Source tab showing the Union and 300,000 rows]
2.	Data Preprocessing	• Unioned train and test files (Specific/Wildcard union) and verified 300,000 rows. • Data types checked: model_year set to Number (whole) → Discrete; sales, range, rating, price fields as numeric measures; market_region, category, battery_type, manufacturer as strings. • manufacturing_country assigned Geographic Role > Country/Region (10 countries mapped). • Null check: annual_sales_units ≈ 1% nulls, customer_rating ≈ 2% nulls; battery_cycle_life and resale_value_pct_3yr also contain nulls. Tableau ignores nulls in SUM/AVG, so values were kept and documented. • Duplicate check: 1 duplicate vehicle_id found out of 300,000 rows – negligible, retained. [Insert screenshot: Data Source grid / data types]
3.	Utilization of Filters	Dashboard filters (applied to all worksheets using this data source): Market Region, Model Year, Battery Type, Category. Worksheet filters: Top 10 Manufacturer by Sum of Annual Sales Units (V2); Top 5 Category (V3). Interactive filter actions: V7 (country map), V4 (battery donut) and V1 (category bars) set to 'Use as Filter' to cross-filter the other visuals. [Insert screenshot: dashboard with filter panel]
4.	Calculation fields Used	1. Total Sales = SUM([Annual Sales Units]) 2. Sales (Millions) = SUM([Annual Sales Units]) / 1000000

		3. Market Share % = SUM([Annual Sales Units]) / MIN({ FIXED : SUM([Annual Sales Units]) }) — FIXED LOD so market share stays relative to all 200 manufacturers when the Top 10 filter is applied. Quick table calculations: Percent of Total (V4 battery share), Percent Difference (V3 growth view). [Insert screenshot: Calculated field editor for Market Share %]
5.	Dashboard design	No of Visualizations / Graphs - 9 (V1–V9): V1 Annual sales by category; V2 Top 10 manufacturers & market share; V3 Sales trend by model year; V4 Battery type share; V5 Average range by category; V6 Range by battery type over years; V7 Manufacturing country map; V8 Customer rating by category; V9 Region × category heatmap. [Insert screenshot: final dashboard]
6	Story Design	No of Visualizations / Graphs - 4 story points – Overview dashboard + one point per persona: Emily Johnson (V1–V3), David Chen (V2, V4, V7, V9), Dr. Sarah Williams (V4, V5, V6, V8); each with a caption stating the key insight. Together the story covers all 9 visualizations. [Insert screenshot: Story view]